

Oscilloscope

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An oscilloscope is a device which is primarily used to graph an electrical signal as it varies over time. A number of us have already seen an oscilloscope in use as an EKG machine. An image of a possible oscilloscope in our labs is Figure 1.

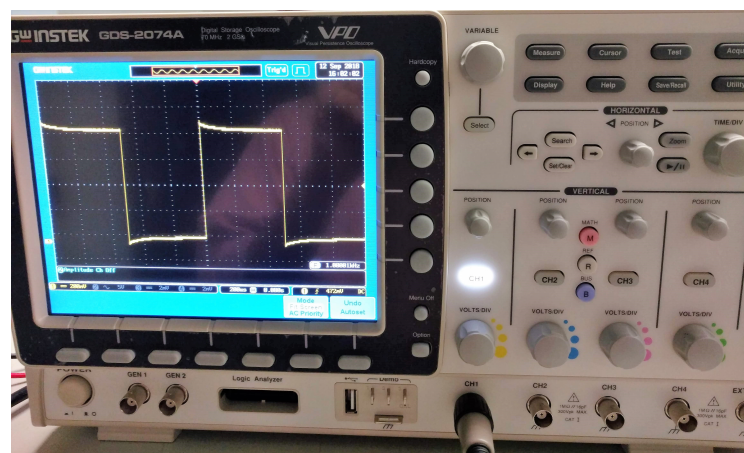


Figure 1: Oscilloscope

The first thing you'll notice about the scope is the screen where the waveform will be displayed. The scope produces a two-dimensional graph with time on the x-axis and voltage on the y-axis. We'll talk more about the display later, but let's first look at how we provide the input signal to the scope.

Probes and Channels

Probes are input devices that route signal from the circuit to the oscilloscope. A typical probe is shown in Figure 2. The signal read by probe is fed into a channel in the oscilloscope through BNC connector. The number of signals an oscilloscope can read simultaneously is equal to the number of channels. The oscilloscopes in the lab have four channels, labeled as *CH1*, *CH2*, *CH3*, and *CH4*.

The other end of the probe has a sharp **tip** which probes into a point on the circuit. It also has a **ground clip** which should be connected to the common ground point of the circuit. There



Figure 2: Oscilloscope Probe



Figure 3: Attenuation probe

is also a switch on the probe as shown in Figure 3. This switch should be set to $1X$ for us, otherwise the amplitude will be reduced.

Displaying a signal

Displaying a signal on the oscilloscope is easy. The ground clip of the probe needs to be connected to the ground point of the circuit and metal tip should be placed at the point where you want to measure voltage. **Note that voltage signals displayed on the oscilloscope are always between a point on the circuit and the ground point. You cannot directly measure the voltage between any two arbitrary points in the circuit using one probe.**

Press the channel button (*CH1*, etc) for the respective channel your probe is connected to. When a channel is activated, its corresponding button will light up. If you want to deselect a channel you can press that channel button again.

Press *Autoset* button in top-right corner and your signal will be displayed on the screen.

Display and Knobs

The display of a scope, as illustrated in Figure 4, contains a lot of information. At the same time, the scope has a number of knobs and controls available as well. It is easy to get overwhelmed by all of this, but we'll discuss the necessary information below:

- **Channel Indicator:** On the very left of the screen, you'll see a number next to the signal being displayed which is channel number of the signal. If the scope has a colored display then the signal will be of the same color as the channel. e.g. the 1 next to sine wave in Figure 4.
- **Horizontal Scale:** If you look at the display, you'll notice a grid, which means that there should be a scale associated with it. The scale for the horizontal axis is displayed in the

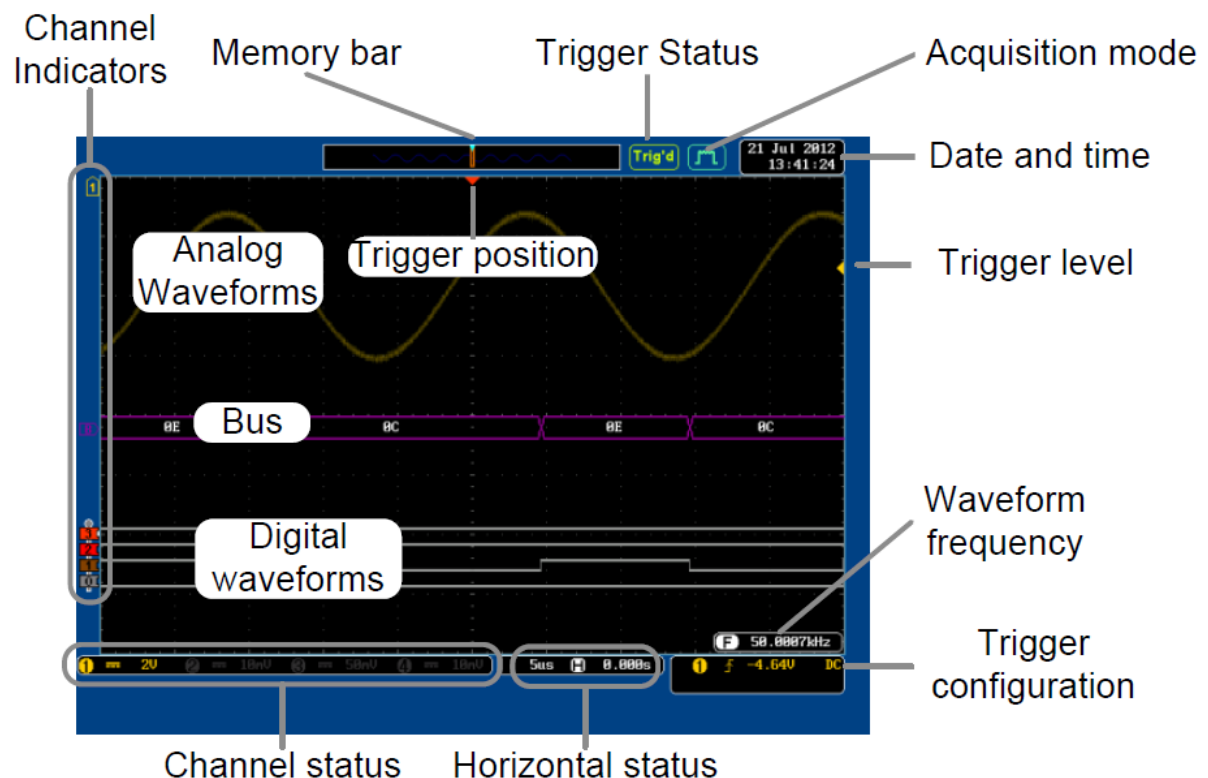


Figure 4: Display of scope

section marked as “Horizontal status” in Figure 4. The $5\mu\text{s}$ in this section indicates that one block of the grid corresponds to $5\mu\text{s}$ on horizontal axis. The scale is automatically set when you press the *Autoset* button, but you can change it or set it manually using the *TIME/DIV* knob.

- **Vertical Scale:** Analogous to the horizontal scale, we have a vertical scale as well. This scale is displayed in the section marked as “Channel status” in Figure 4. The 2V in this section indicates that one block of the grid corresponds to 2V on vertical axis. Again, the scale is automatically set when you press the *Autoset* button, but you can change it or set it manually using the *VOLTS/DIV* knob of the respective channel.
- **Frequency:** The oscilloscope computes the frequency of the inputted signal and displays it in the bottom right corner as shown in Figure 4.
- **Coupling:** In the “Channel status” section in Figure 4, you’ll see a --- symbol next to the vertical scale. This indicates that the scope has been set to DC coupling. In this mode, the entire signal appears on the screen. Contrastingly, in AC coupling only the AC portion of signal is displayed or the signal is centered around zero. One can switch between the coupling modes by pressing the *CH* button and then repeatedly pressing *Coupling* button.

at bottom of screen to toggle through. **Make sure that you're in DC coupling mode for this lab.**

Measurements

The oscilloscope also has the feature to automatically measure and display common desired parameters of a signal. This includes measuring the amplitude, time-period, frequency, duty cycle, high level, low level, etc. To add an automatic measurement item:

1. Press *Measure* key at the top.
2. Press *Add Measurement* from bottom menu.
3. Set the source by pressing *Source 1* in the side menu.
4. Choose either *V/I* or *Time* from the side menu and choose the type of measurement.
5. All measurements will be displayed in a window on bottom of screen. The source of measurement is shown as channel number next to it.

Resources

You can find out more about oscilloscopes by reading its user manual or at

- <https://learn.sparkfun.com/tutorials/how-to-use-an-oscilloscope>
- <https://www.allaboutcircuits.com/worksheets/basic-oscilloscope-operation/>